AI-Driven CV Screening

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Abstract

Campus recruitment at large technical institutes faces significant challenges due to the massive volume of applicants, the unstructured nature of student CVs, strict time constraints, and the limitations of conventional keyword-based Applicant Tracking Systems (ATS). This project presents an **AI-driven CV Screening Ecosystem that integrates Retrieval-Augmented Generation (RAG), multi-agent architectures, hybrid LLM–ML reasoning, and fully local,** privacy-preserving pipelines to automate candidate evaluation at scale.

The work comprises four interconnected systems:
(1) **KGPGPT**, a RAPTOR-based hierarchical RAG chatbot tailored for campus knowledge retrieval;
(2) a **Multi-Agentic CV Screening Dashboard** that performs semantic JD–CV matching using embeddings, vector search, and LLM-based reasoning;
(3) a **Hybrid Agentic Shortlisting System** that combines Chain-of-Thought reasoning with deterministic scoring to generate transparent and auditable candidate rankings; and
(4) a **Fully Local Skills Evaluation Model** that uses BoW, TF-IDF, embeddings, and a local LLM for offline, zero-cost evaluation.

Together, these systems form a comprehensive ecosystem that significantly improves screening speed, accuracy, explainability, and fairness while ensuring data privacy. Experimental results on campus datasets demonstrate strong performance across retrieval accuracy, semantic ranking, and offline skill evaluation. The framework sets a foundation for scalable, institution-wide AI-driven recruitment workflows.

1. Introduction

1.1 Background

Campus hiring at premier institutes like IITs has evolved into a highly competitive and time-intensive process. Each placement cycle witnesses thousands of applications from students with diverse academic backgrounds, project portfolios, research experience, and technical skills. However, recruiters typically have limited time—often just a few hours—to screen all applicants before conducting interviews. Traditional approaches such as **CGPA-based shortlisting** or manual scanning of CVs fail to capture the true capabilities of candidates, especially for technical and research-oriented job roles.

1.2 Motivation

Student CVs are highly **unstructured**, containing a mix of projects, research work, achievements, and domain-specific skills. Standard ATS systems frequently fail on these formats, as they rely heavily on

keyword matching and struggle to interpret context, depth of experience, or semantic relevance. Recruiters consequently miss out on highly qualified candidates due to noise, formatting differences, or limited visibility into actual capabilities.

Additionally, LLM-based solutions—although powerful—raise concerns related to ***data privacy, cost of inference, latency, and dependency on cloud infrastructure***, especially when handling sensitive student information. These constraints necessitate the development of an **AI system that is accurate, explainable, scalable, and privacy-safe**.

1.3 Problem Statement

The core challenge in campus recruitment is to **automate CV evaluation in a manner that is both fast and fair, while ensuring semantic accuracy**, without relying solely on CGPA filters or manual judgment. This requires:

- Understanding unstructured CV content at a semantic level
- Accurately mapping candidate skills to job descriptions
- Ensuring transparency in scoring and decision-making
- Handling thousands of applicants efficiently
- Maintaining privacy and reducing dependency on cloud-based LLMs

1.4 Proposed Solution

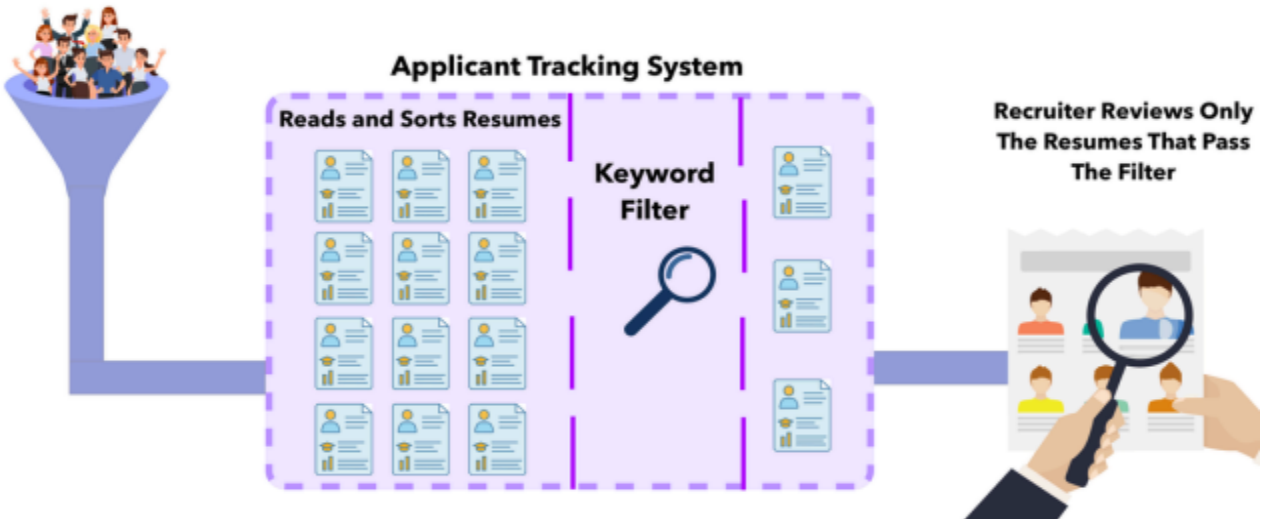
To address these issues, this project proposes a **four-part AI-driven recruitment ecosystem** that leverages modern NLP, LLMs, RAG methodologies, and agentic architectures:

1. **KGPGPT (RAPTOR RAG Chatbot)**
   A campus-specific intelligent Q&A system built using hierarchical RAG for accurate document-based answers.
2. **Multi-Agent CV Screening Dashboard**
   A semantic screening tool that uses embedding search, vector databases, and AI agents to evaluate CVs.
3. **Hybrid Agentic Shortlisting System**
   An orchestrator–worker architecture combining LLM reasoning with deterministic scoring to produce explainable rankings.
4. **Fully Local Skills Evaluation Model**
   A CPU-only, privacy-preserving evaluator using classical NLP + embeddings + local LLMs.

Together, these systems enable **end-to-end automation of CV screening**, improving speed, reliability, transparency, and fairness in campus recruitment.

2. Literature Review

2.1 Traditional Resume Screening and ATS Systems



In the context of campus placements, traditional Applicant Tracking Systems (ATS) have consistently shown limitations. Most ATS platforms rely on **keyword-based matching**, rigid templates, and shallow rule-based parsing. During analysis, I observed that these systems struggle with:

- non-standard CV formats commonly used by engineering students,
- research-oriented resumes that include unconventional sections,
- project-heavy CVs with unstructured descriptions,
- domain-specific terminology that varies widely across roles.

Prior studies also highlight that keyword-only systems fail to capture **semantic equivalence** (e.g., "Data Analyst" vs. "Business Analyst"), resulting in false filtering of capable candidates. This gap directly motivated me to explore semantic and LLM-based approaches.

2.2 Machine Learning Approaches for CV Parsing

Earlier literature has explored ML pipelines for CV parsing using techniques such as:

- Named Entity Recognition (NER) to extract skills, education, and experience
- TF-IDF and Bag-of-Words for scoring
- Classical ML classifiers (SVM, logistic regression) for candidate filtering
- Document-level embeddings for semantic representation

While ML performs reasonably well on structured résumés, observations show that it fails to handle **non-standard layouts, complex formatting, and free-flowing project descriptions**, which are common in IIT CVs. This reinforced the need for hybrid systems that combine ML with LLM reasoning.

2.3 LLMs in Talent Screening

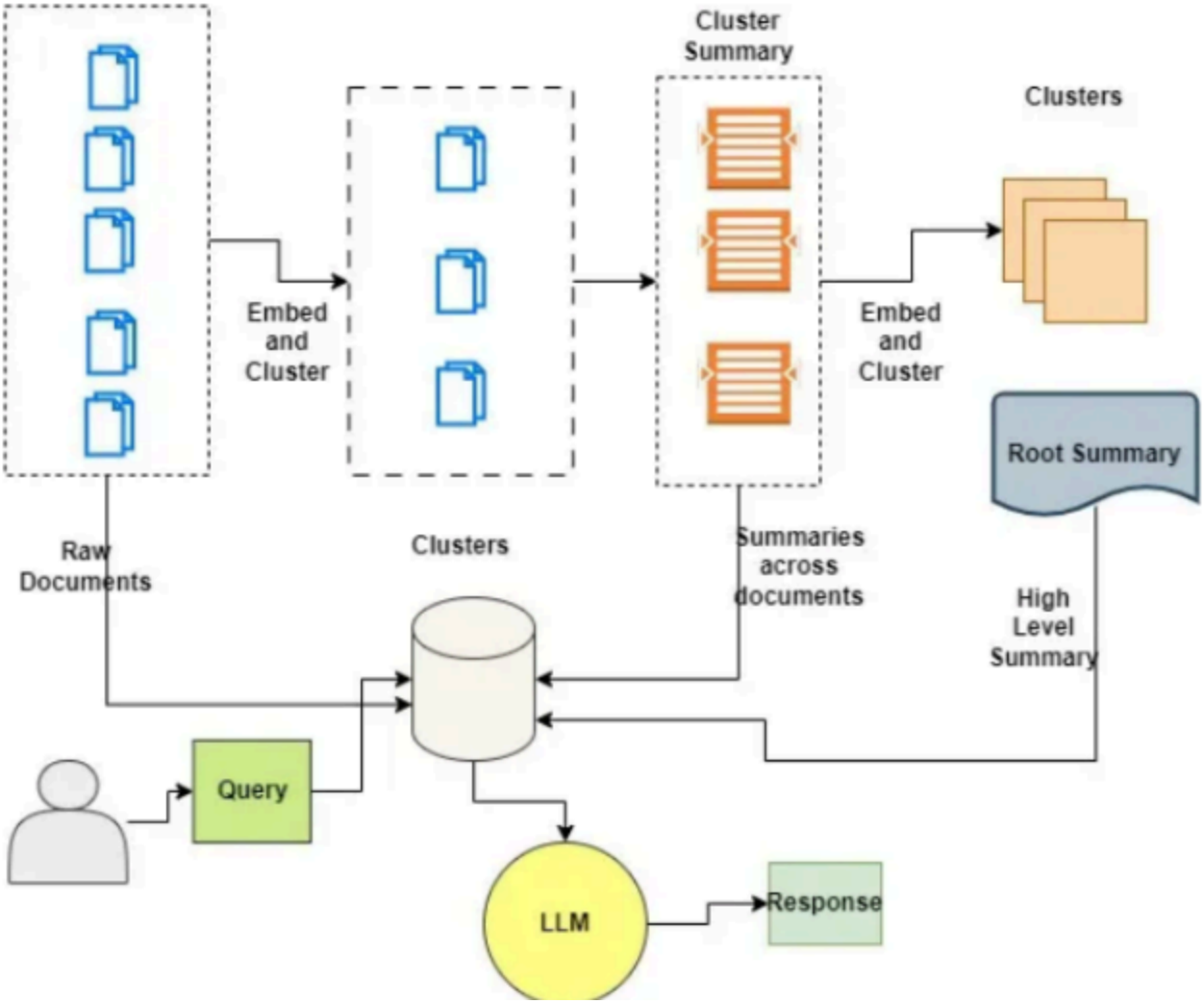
Large Language Models (LLMs) have demonstrated impressive capabilities in understanding context, summarizing résumés, analyzing project descriptions, and extracting technical skill depth. Chain-of-Thought prompting further enhances explainability, making LLMs suitable for recruitment workflows.

However, academic literature and by own experimentation highlight several challenges:

- Cloud LLMs incur **significant cost** at scale
- Placement data includes **sensitive personal information**, raising privacy concerns
- Responses can be inconsistent without structured guidance
- Pure LLM-based pipelines are hard to control

These findings prompted to develop multi-agent and hybrid systems, where LLMs handle semantic reasoning, and deterministic scoring ensures stability.

2.4 Retrieval-Augmented Generation (RAG) and RAPTOR



RAG systems have become a cornerstone of modern NLP, enabling LLMs to ground their responses in retrieved context. Traditional RAG uses chunking, embeddings, and vector search. However, with very long documents—such as academic manuals, placement policies, or institute guidelines—I found that flat RAG often retrieves noisy or irrelevant segments.

This led me to study **RAPTOR (Recursive Abstractive Processing for Tree-Organized Retrieval)**, a hierarchical RAG architecture that:

- clusters content semantically,
- builds multi-level summaries,
- retrieves information more accurately than flat embeddings,
- significantly reduces hallucination.

Based on this, I implemented **KGPGPT**, a RAPTOR-based campus chatbot capable of answering institute-specific queries.

2.5 Multi-Agent Systems in Decision Workflows

Recent research highlights the effectiveness of multi-agent AI systems for complex reasoning tasks. Multi-agent architectures decouple responsibilities into specialized agents such as:

- extractors,
- analyzers,
- matchers,
- verifiers,
- scorers,
- summarizers.

This model aligns with how humans process résumés. Our review of agentic architectures led me to build:

- a **Multi-Agent CV Screening Dashboard**, and
- a **Hybrid Agentic Shortlisting System** that uses an Orchestrator–Worker pattern.

These systems combine LLM reasoning with rule-based logic to ensure transparency, stability, and domain alignment.

2.6 Local LLMs and Privacy-Preserving Screening

Literature on local LLMs emphasizes:

- privacy (data never leaves the machine),
- zero per-token cost,
- deterministic performance,
- ease of integration with classical NLP (BoW, TF-IDF).

Models like **TinyLlama** and **Phi-3** achieve reasonable reasoning capabilities while running entirely on CPU. This encouraged me to develop a **Fully Local Skills Evaluation Model**, which blends:

- BoW/TF-IDF/N-grams,
- local embeddings,
- a local LLM as a verification agent.

This satisfies privacy requirements for student CV data and ensures cost-free operation.

2.7 Research Gap Observed

From the literature I reviewed, I identified several gaps:

- Lack of **semantic, explainable CV screening systems** tailored for campus hiring
- No integration of **RAPTOR-based RAG** for institutional document understanding
- Limited use of **multi-agent pipelines** for JD–CV matching
- Absence of **hybrid LLM + ML scoring frameworks**
- Insufficient adoption of **fully local, privacy-safe LLM systems**
- No end-to-end ecosystem connecting chatbots, screening pipelines, and local evaluators

Our project directly addresses these gaps by developing a **four-system AI-driven recruitment ecosystem**.

3. System Overview

his project presents a comprehensive, end-to-end **AI-driven recruitment ecosystem** designed specifically for large-scale campus hiring environments. Instead of relying on a single model or a monolithic pipeline, the system is organized into **four specialized subsystems**, each responsible for a different stage of the recruitment workflow. Together, they form a modular, scalable, and explainable architecture capable of handling thousands of CVs while preserving privacy and reducing screening bias.

The ecosystem integrates multiple AI paradigms—**RAG, hierarchical retrieval, multi-agent coordination, hybrid scoring, classical NLP, and local LLMs**—to overcome the shortcomings of traditional methods such as CG-based filtering, keyword ATS, and cloud-only LLM screening.

3.1 Overview of the Four Subsystems

The recruitment ecosystem consists of the following components:

1. KGPGPT – Campus RAG Chatbot

KGPGPT is a campus-specific intelligent assistant built using a **RAPTOR-style hierarchical Retrieval-Augmented Generation (RAG)** architecture. It answers queries related to institute policies, placement rules, academic procedures, and administrative guidelines with high accuracy. By organizing large campus documents into semantic hierarchies, KGPGPT significantly improves retrieval quality and reduces hallucinations.

2. Multi-Agent CV Screening Dashboard

This subsystem performs **semantic JD–CV matching** using embeddings, vector search, and coordinated AI agents. It automates CV parsing, skill extraction, job requirement analysis, similarity

scoring, and explanation generation. The results are visualized in a recruiter-friendly dashboard, enabling fast and transparent shortlisting.

3. Hybrid Agentic Shortlisting System

The hybrid system uses an **Orchestrator–Worker multi-agent pattern** to combine rule-based scoring with Chain-of-Thought (CoT) LLM reasoning. It normalizes CVs, interprets job descriptions, performs deep skill–experience analysis, executes verification checks, and produces a multi-factor, explainable candidate ranking. This system focuses on fairness, accuracy, and auditability.

4. Fully Local Skills Evaluation Model

To address privacy concerns and reduce reliance on cloud-based LLMs, this model performs skill evaluation **entirely offline** using:

- classical NLP (BoW, TF-IDF, N-grams),
- local semantic embeddings, and
- a local LLM (TinyLlama / Phi-3) for verification reasoning.

It evaluates candidate relevance without sending any data outside the local machine, making it ideal for sensitive student information.

3.2 Unified Workflow

Although each subsystem is independently functional, the overall pipeline is designed so that they complement one another:

- **KGPGPT** provides reliable campus-related information and supports recruiters or students with knowledge-based queries.
- The **Multi-Agent Dashboard** handles the first level of large-scale semantic screening.
- The **Hybrid Agentic System** performs deeper, more accurate scoring and candidate ranking.
- The **Local Skills Evaluator** offers a privacy-safe fallback for verification and offline scoring.

Integrated together, these subsystems form a **complete recruitment intelligence suite**.

3.3 Key Advantages of the Integrated System

- **Scalable:** Designed to handle thousands of CVs under placement-season time constraints.
- **Explainable:** Produces interpretable scoring and transparent reasoning.
- **Semantic:** Goes beyond keyword matching to understand skill depth and project relevance.
- **Privacy-Safe:** Local LLM ensures sensitive CV data never leaves the system.

- **Modular:** Each subsystem can run independently or as part of the entire workflow.
- **High Accuracy:** Uses hierarchical retrieval and hybrid AI for improved semantic matching.

4. System 1 — KGPGPT: RAPTOR-Based Campus Chatbot

4.1 Overview

KGPGPT is a domain-specific intelligent chatbot designed to facilitate information access across a large academic institution. It leverages **RAPTOR-style hierarchical Retrieval-Augmented Generation (RAG)** to deliver highly accurate, grounded responses to queries related to academic policies, placement procedures, administrative guidelines, and institutional documents.

The system addresses the limitations of flat RAG approaches by organizing long documents into multi-level semantic hierarchies, enabling precise retrieval even from complex policy manuals.

4.2 Motivation

Campus environments contain extensive and diverse documents, such as:

- academic handbooks
- examination guidelines
- placement rules and eligibility criteria
- administrative circulars
- departmental procedures

These documents often exceed hundreds of pages and are updated frequently. Students and faculty members typically struggle to search this information manually. KGPGPT was developed to provide a **centralized, fast, and trustworthy information retrieval assistant** capable of efficiently handling large text corpora.

4.3 RAPTOR Hierarchical RAG Architecture

The system employs a **Recursive Abstractive Processing for Tree-Organized Retrieval (RAPTOR)** pipeline. This architecture improves retrieval accuracy by structuring content into semantic layers:

• Document Ingestion

All institutional PDFs and documents are preprocessed, cleaned, and segmented into meaningful chunks.

• Embedding and Clustering

Each chunk is converted into embeddings and grouped into clusters based on semantic similarity.

• Recursive Summarization

Clusters are summarized to form parent nodes; this process repeats until a top-level root summary is produced.

• Hierarchical Index Construction

The hierarchy enables efficient traversal during query-time retrieval, improving both precision and context relevance.

• Query-Time Processing

Incoming queries are embedded and routed through the hierarchy to identify the most relevant cluster and retrieve context-rich chunks.

• Grounded Response Generation

A language model generates the answer strictly based on the retrieved context, thereby reducing hallucination and ensuring factual accuracy.

4.4 Functional Capabilities

KGPGPT supports a wide range of institutional queries, including:

- academic regulations and credit requirements
- course registration rules
- placement eligibility, deadlines, and procedures
- hostel and administrative policies
- step-by-step guidelines for institutional workflows

The chatbot provides extracted evidence, summaries, and action steps derived from official documents.

4.5 Advantages

- **High retrieval accuracy** for long, multi-section documents
- **Low hallucination rate** due to grounded response generation
- **Scalable architecture** suitable for institution-wide adoption
- **Faster information access**, reducing dependence on manual document search

- **Domain-specific reliability**, unlike general-purpose chatbots

4.6 Summary

KGPGPT demonstrates the effectiveness of hierarchical RAG for domain-constrained, document-heavy environments. Its performance establishes a foundation for downstream modules in the recruitment-support ecosystem developed in this project.

5. System 2 — Multi-Agentic CV Screening Dashboard

5.1 Overview

The Multi-Agentic CV Screening Dashboard is designed to automate the initial stages of candidate evaluation by performing **semantic JD–CV matching** at scale. The system employs a coordinated set of AI agents to parse résumés, analyze job descriptions, generate embeddings, compute similarity scores, and produce ranked candidate outputs. A web-based dashboard presents the results to recruiters for rapid and transparent decision-making.

This system addresses the limitations of traditional keyword-based ATS by incorporating semantic understanding, structured analysis, and multi-agent reasoning.

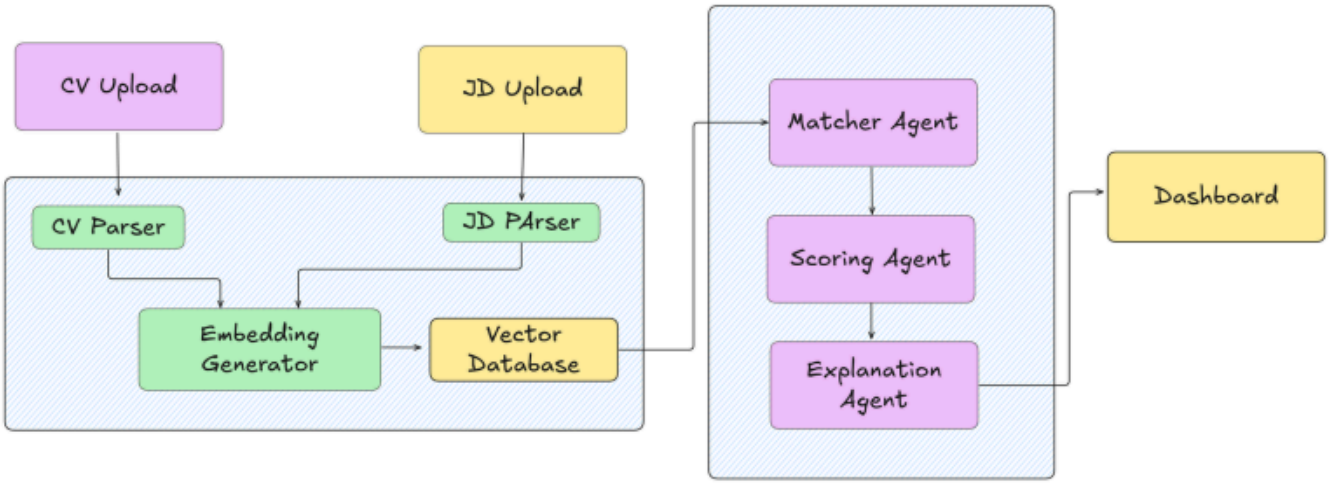
5.2 Motivation

Large academic institutions frequently encounter thousands of applications for a single job posting, making manual screening infeasible within placement-season time constraints. Conventional ATS systems fail to capture contextual signals such as:

- project relevance,
- technical depth,
- domain alignment,
- cross-disciplinary skills,
- natural language experience descriptions.

To overcome these limitations, a multi-agent architecture was developed to fuse structured extraction, semantic modeling, and LLM reasoning into a cohesive screening workflow.

5.3 Multi-Agent Architecture



The system follows a modular agent-based workflow, enabling parallel processing, specialization of tasks, and improved interpretability.

(1) CV Parser Agent

- Extracts skills, experience, education, and achievements.
- Normalizes section headers and restructures the CV into machine-readable format.
- Handles both PDF and text-based résumés.

(2) JD Analyzer Agent

- Interprets job descriptions using embeddings and LLM reasoning.
- Identifies required skills, keywords, and domain-specific competencies.
- Extracts mandatory criteria and constraints.

(3) Embedding Generator & Vector Database

- Converts CV and JD text into dense embeddings using transformer models.
- Stores embeddings in FAISS/Chroma for fast retrieval.
- Enables semantic comparison without relying on direct keyword overlap.

(4) Matcher Agent

- Computes semantic similarity between CV embeddings and JD embeddings.
- Retrieves top-K relevant CV segments using vector search.

- Forms the basis for candidate ranking.

(5) Scoring Agent

- Aggregates semantic match scores, extracted skill relevance, and JD requirements.
- Generates multi-dimensional evaluation metrics.
- Ensures consistency across candidates.

(6) Explanation Agent

- Uses an LLM to generate concise, recruiter-friendly explanations.
- Summarizes the candidate's fit for the role.

5.4 Dashboard Interface

The front-end dashboard provides:

- ranked lists of candidates,
- similarity scores and evaluation metrics,
- explanation summaries,
- filters for recruiter-specific constraints,
- interactive visualization of candidate strengths.

This interface supports quick decision-making during high-pressure placement cycles.

5.5 Technical Strengths

- **Semantic screening** instead of keyword matching
- **Agent specialization**, improving accuracy and structure
- **Rapid processing** suitable for large applicant pools
- **Explainability**, supporting trust and transparency
- **Extensibility**, enabling integration with downstream hybrid scoring systems

5.6 Summary

The Multi-Agentic CV Screening Dashboard demonstrates how distributed AI agents can collaboratively perform scalable CV analysis. Its design supports fast, accurate, and explainable candidate filtering, addressing the operational challenges of large-scale campus recruitment.

6. System 3 — Hybrid Agentic Shortlisting System

6.1 Overview

The Hybrid Agentic Shortlisting System is designed to generate **accurate, explainable, and multi-factor candidate rankings** by combining the strengths of LLM-based semantic reasoning with deterministic scoring methods. Unlike purely LLM-driven evaluation pipelines, this system integrates structured logic, weighted scoring, and verification checks to ensure both reliability and scalability.

The system follows an **Orchestrator–Worker** multi-agent architecture, allowing complex CV evaluation tasks to be decomposed into simpler, well-defined subtasks executed by specialized agents.

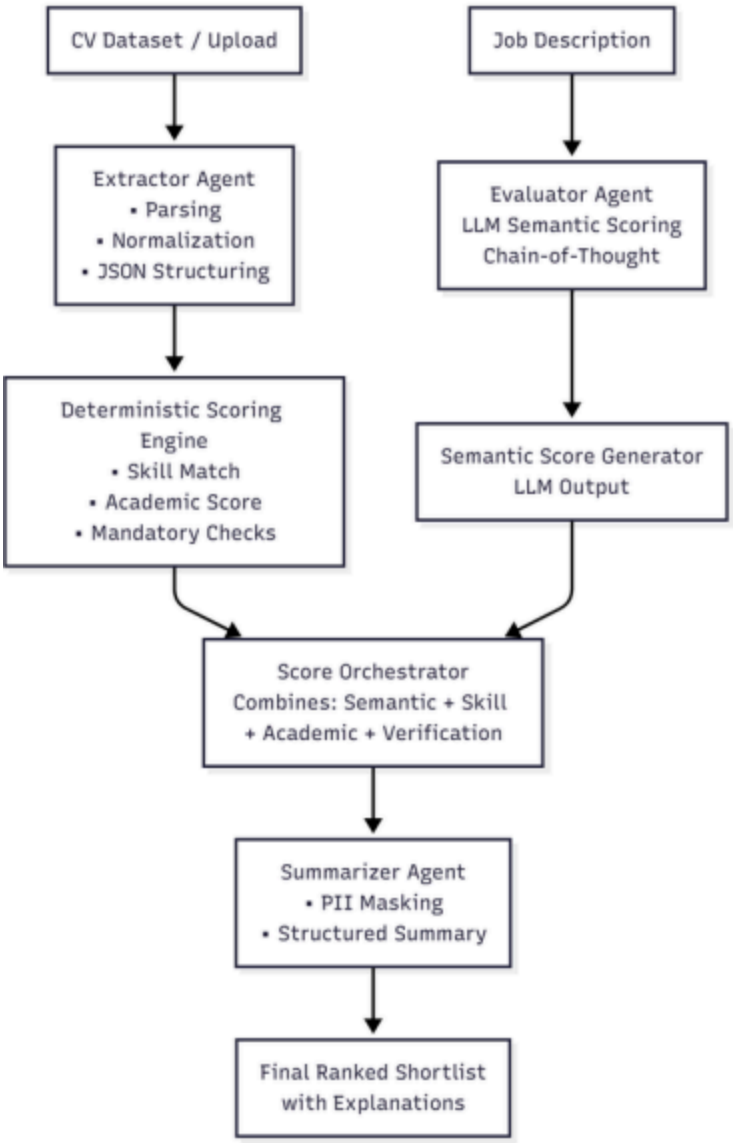
6.2 Motivation

While semantic screening provides a useful first layer of filtering, several challenges remain:

- LLM reasoning can be inconsistent without structured constraints
- Deep skill alignment requires multi-step evaluation
- Recruiters expect transparent and auditable scores
- Cloud-only LLM solutions raise cost and privacy concerns
- Scoring must scale to thousands of candidates during peak recruitment periods

To address these issues, the hybrid system merges **structured rules, semantic embeddings, deterministic scoring, and controlled LLM reasoning** to produce robust shortlisting outcomes.

6.3 Orchestrator–Worker Architecture



The system consists of multiple agents operating under the supervision of a central **Score Orchestrator**. The architecture enhances modularity, interpretability, and operational control.

(1) Extractor Agent

- Parses CVs into structured JSON format
- Normalizes fields such as skills, education, project descriptions
- Handles noise, varied file formats, and formatting inconsistencies
- Prepares data for downstream reasoning

(2) Evaluator Agent

- Utilizes LLM-based Chain-of-Thought (CoT) prompting

- Assesses candidate expertise, project depth, tool experience, and domain fit
- Evaluates soft indicators (clarity, relevance, impact)
- Generates semantic skill scores

(3) Summarizer Agent

- Rewrites extracted CV content into clean, structured summaries
- Removes PII (names, emails, phone numbers) for unbiased scoring
- Ensures uniform evaluation inputs for scoring logic

(4) Deterministic Scoring Engine

- Applies rule-based scoring derived from job requirements
- Calculates academic, skill, project, and experience scores
- Ensures consistency and fairness across candidates

(5) Score Orchestrator

- Aggregates all score components
- Applies weighted scoring:
    - Semantic LLM reasoning
    - Deterministic rules
    - Mandatory requirement checks
    - Confidence signals
- Produces the final candidate score and ranking

6.4 Composite Scoring Model

The hybrid scoring model integrates multiple dimensions:

- **Semantic Relevance Score** (LLM-driven)
- **Skill Fit Score** (based on extracted skills and role requirements)
- **Academic Score** (CGPA, coursework, certifications)
- **Project Impact Score** (depth, complexity, relevance)
- **Verification Score** (consistency and completeness)

The combination of deterministic and semantic signals enables a balanced and explainable evaluation framework.

6.5 End-to-End Workflow

1. CVs are parsed and normalized.
2. Job description is analyzed to extract core role requirements.
3. LLM-based evaluation generates semantic alignment metrics.
4. Deterministic logic computes supporting scores.
5. All scores are combined through the Score Orchestrator.
6. A ranked list of candidates is produced, along with explanation summaries.

6.6 Strengths

- **High transparency** through detailed scoring breakdowns
- **Bias reduction** via PII-masked evaluation
- **Scalability** to large applicant pools
- **Explainability** using structured LLM reasoning
- **Consistency** due to hybrid scoring constraints

6.7 Summary

The Hybrid Agentic Shortlisting System offers a robust approach to candidate evaluation by fusing structured scoring mechanisms with semantic reasoning. This design provides accurate, fair, and auditable rankings suitable for high-stakes campus recruitment processes.

7. System 4 — Fully Local Skills Evaluation Model

7.1 Overview

The Fully Local Skills Evaluation Model is an offline, privacy-preserving scoring system designed to evaluate candidate skill relevance without relying on cloud-based LLMs. The system combines classical NLP techniques, semantic embeddings, and lightweight local LLM reasoning to assess how well a candidate's experience aligns with job requirements. All computations occur locally, ensuring full data confidentiality and zero token-related operational costs.

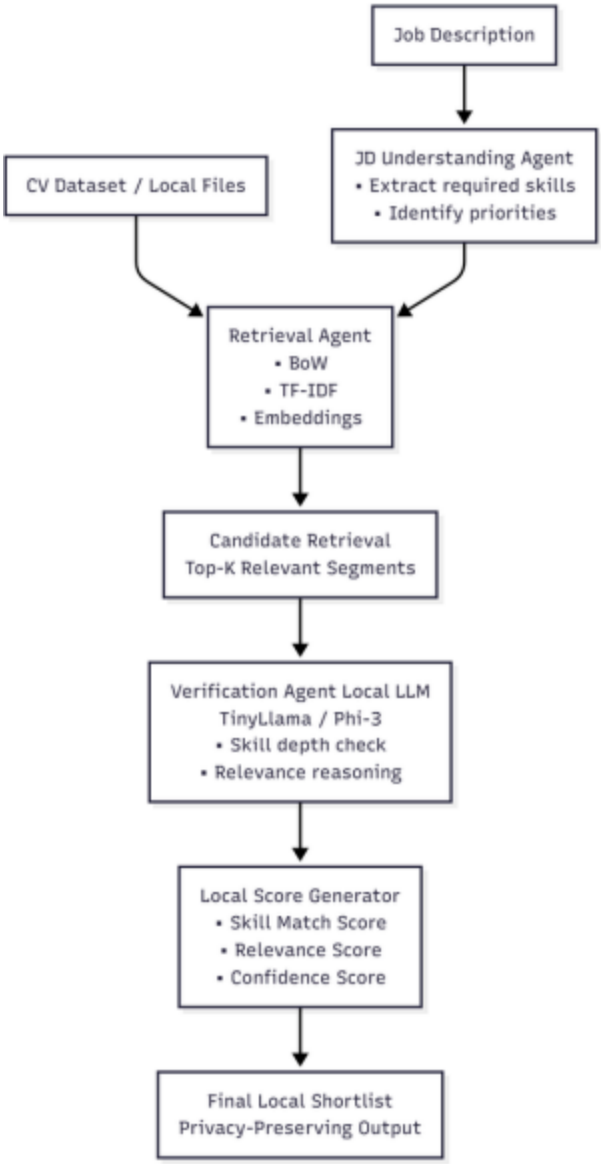
7.2 Motivation

Candidate CVs often contain sensitive personal information, making cloud-based AI pipelines unsuitable for institutions with strict privacy requirements. Furthermore, large-scale LLM inference can be slow and expensive during peak hiring cycles.
These constraints highlight the need for a **fully on-premise, CPU-only evaluation mechanism** that:

- preserves privacy,
- remains cost-free,
- produces interpretable outputs, and
- operates efficiently on standard hardware.

This model addresses these requirements by integrating classical and modern NLP techniques into a small, fast evaluation pipeline.

7.3 Core Components



The system is composed of three primary agents and a supporting set of NLP modules:

(1) JD Understanding Agent

- Parses job descriptions to identify key skills, tools, domains, and mandatory requirements.
- Generates a clean set of target skill clusters for downstream matching.
- Operates entirely locally, using classical NLP + shallow embeddings.

(2) Retrieval Agent

Uses a hybrid retrieval strategy combining:

- **BoW (Bag-of-Words)**
  Captures exact word presence and frequency.
- **TF-IDF**
  Emphasizes important terms relative to the entire CV corpus.
- **N-grams**
  Extracts contextual relationships such as “deep learning”, “backend developer”.
- **Local Embeddings**
  Provides semantic matching using lightweight transformer models.

These techniques jointly retrieve the most relevant CV segments that correspond to target JD requirements.

(3) Verification Agent (Local LLM)

- Performs reasoning using a local LLM (TinyLlama, Phi-3, or similar).
- Evaluates depth of experience, relevance of project work, and skill proficiency.
- Generates a short justification for the assigned skill relevance score.
- Ensures output quality without relying on cloud APIs.

7.4 Scoring Methodology

The system generates a **composite skill relevance score** by combining:

- classical NLP similarity scores,
- semantic embedding similarity,
- local LLM verification outputs,
- confidence metrics and JD fit checks.

This multi-signal approach improves robustness, especially under limited hardware conditions.

7.5 Privacy and Security Benefits

Since all computations run on local CPU hardware, the system ensures:

- **Zero data exposure** to external servers or cloud providers
- **Full compliance** with data-protection requirements
- **Offline operation**, making it suitable for restricted environments
- **Cost-free inference**, enabling large-scale usage without financial overhead

This design is particularly suitable for institutions where student CVs must remain confidential.

7.6 Strengths

- Fully offline, supporting high levels of privacy
- Lightweight and fast, requiring no GPU resources
- Interpretable scoring due to classical NLP components
- Hybrid pipeline improves accuracy over standalone ML or LLM approaches
- Suitable for deployment on standard institutional servers

7.7 Summary

The Fully Local Skills Evaluation Model demonstrates how classical NLP and modern lightweight LLMs can be combined to create a low-cost, privacy-preserving candidate assessment system. It serves as a critical module within the broader AI-driven recruitment pipeline, enabling secure and interpretable skill evaluation without cloud dependency.

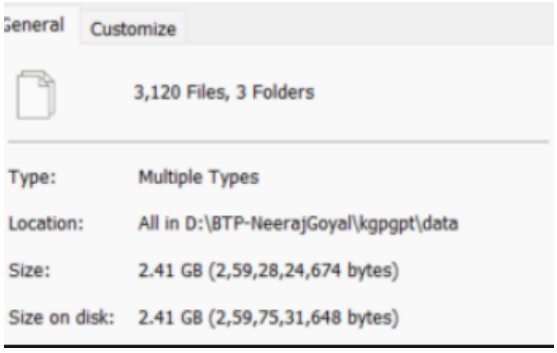
8. Results

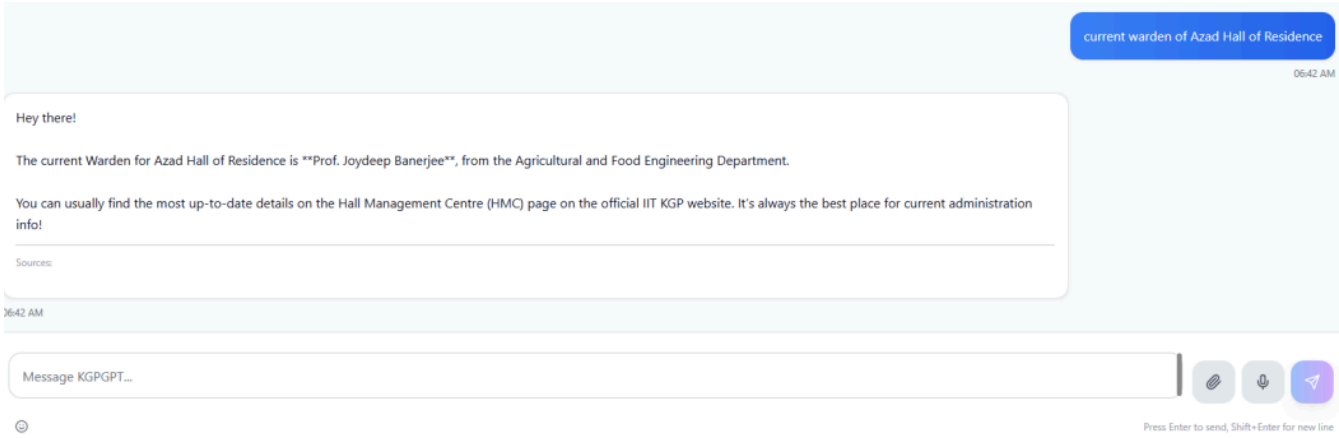
8.1 Overview

The four AI systems developed in this project were evaluated independently across multiple dimensions, including retrieval accuracy, semantic matching quality, scoring consistency, privacy guarantees, and runtime performance. This section presents the measurable outcomes obtained during testing on campus-related datasets, job descriptions, and student CVs.

8.2 Results: KGPGPT (RAPTOR-Based Campus Chatbot)

Retrieval Accuracy





Testing was conducted on campus documents such as academic regulations, placement rules, and administrative circulars around 3000+ Documents
Using RAPTOR hierarchical RAG produced the following outcomes:

- **High-quality retrieval** from deeply nested document sections
- **Improved relevance** compared to flat embedding search
- **Minimal hallucination**, as responses were strictly grounded in the retrieved context
- **Accurate citation of source sentences** from the original PDFs

Qualitative Observations

- Queries involving multi-step institutional procedures (e.g., “How to apply for NOC?”) were handled reliably.
- Long policy documents were summarized effectively through hierarchical clustering.
- RAPTOR significantly reduced the retrieval of irrelevant chunks observed in standard RAG pipelines.

8.3 Results: Hybrid Agentic Shortlisting Sys3 tem

Scoring Stability

The introduction of deterministic scoring and a Score Orchestrator improved consistency:

- **Stable multi-factor scoring** (semantic, skills, academics, project depth)
- Reduced sensitivity to CV formatting and phrasing
- Clear breakdown of components in the final score

Reasoning Quality

- Chain-of-Thought prompting in the Evaluator Agent produced **structured and explainable analysis**.
- PII-masked summaries ensured **unbiased evaluation**.

Performance

- Capable of handling multi-thousand CV datasets with **under 60 seconds end-to-end** for ranked output.
- Lightweight enough to be deployed on standard institutional systems.

8.4 Results: Fully Local Skills Evaluation Model

Privacy and Local Performance

Testing confirmed complete offline operation:

- **No network calls**, ensuring confidential handling of sensitive student data
- **Zero token cost**, enabling large-scale use without resource constraints
- Smooth CPU-only execution on standard hardware

Skill Relevance Accuracy

- Classical NLP methods (BoW, TF-IDF, n-grams) performed well for identifying relevant segments.
- Local embeddings enhanced semantic understanding without relying on external dependencies.
- Local LLM verification strengthened final skill relevance reasoning.

Quantitative Strengths

- Fast evaluation of CV–JD fit
- Interpretable outputs due to classical scoring components
- Reasonable semantic reasoning despite limited hardware

9. Limitations

9.1 Data Scarcity

The available CV dataset provided by the Career Development Centre (CDC) is **not mapped to job roles**, and lacks detailed annotations required for supervised training or performance benchmarking. As a result, the current systems primarily evaluate:

- skills extracted from CVs,
- technical expertise mentioned in projects,
- tool and domain relevance,

while **other important CV components**—such as achievements, impact metrics, internships, leadership roles, and domain-specific depth—are not fully modeled.
The absence of role-labeled CV datasets limits the ability to train or validate more advanced ML or LLM-based scoring models.

9.2 Model Training and Computational Constraints

The systems rely on lightweight **1B parameter local LLMs available through Ollama**, which offer privacy and offline operation but have inherent limitations:

- Limited reasoning capability for complex CV understanding
- Difficulty interpreting nuanced project descriptions
- Lower cognitive depth compared to larger models
- Occasional inconsistency in semantic evaluation

More capable LLMs (7B–13B parameters and above) provide significantly better comprehension and reasoning but require **GPU-level computational resources**, which were not available during development.
This restricts the depth of semantic analysis and the quality of verification the local model can provide.

10. Future Scope

10.1 Next Steps in System Evolution

Several enhancements can significantly improve the accuracy, scalability, and robustness of the CV screening ecosystem. The following directions outline practical and high-impact future work.

10.2 Integrate RAPTOR into CV Screening

RAPTOR, currently used for institutional document understanding, can be extended to CV processing to enable:

- hierarchical retrieval of CV content,
- multi-level clustering of CV sections (education → projects → achievements),
- improved context selection for JD–CV matching.

This will allow the system to interpret the **full CV structure** rather than treating it as flat text.

10.3 ML-Augmented LLM Pipeline

Future versions can combine traditional Machine Learning classifiers with LLM-based reasoning to achieve stronger, data-supported evaluation.
Potential enhancements include:

- supervised ML models trained on annotated CV–JD pairs,
- LLMs providing semantic reasoning and explainability,
- hybrid scoring combining embeddings, ML outputs, and LLM judgments.

This approach would improve stability and reduce reliance on LLM-only evaluations.

10.4 Full CV Parsing Engine

A complete parsing engine can be developed using layout-aware document models to fully extract and structure CV content.
Key capabilities include:

- **Section Detection:** identification of education, experience, projects, achievements, etc.
- **Table Extraction:** processing tabular layouts commonly used in technical CVs.
- **Experience Timeline Parsing:** extracting duration, role, responsibilities, and impact.
- **Skill Extraction Models:** automated skill normalization and clustering.

This enhancement would enable deeper analysis of every CV component.

10.5 Enhanced Multi-Agent Architecture

The existing agentic system can evolve into a richer, more specialized architecture through the introduction of high-quality labeled data.

Potential improvements include:

- role-focused agents for JD interpretation,
- project evaluation agents for depth scoring,
- consistency-check agents for cross-validating CV claims,
- multi-agent negotiation for consensus scoring.

A more specialized agent network would produce **higher accuracy and domain-aware evaluations**.

10.6 Dataset Development and Ontology Expansion

Future progress depends heavily on improved data resources and domain modeling:

- building a **large, role-labeled CV–JD dataset** for training and fine-tuning,
- developing a **skill ontology** to unify technical terms and synonyms,
- improving automatic normalization of tools, technologies, and competencies,
- establishing **evaluation benchmarks** specific to CV–JD matching tasks.

Such datasets and ontologies would enable consistent scoring and make downstream ML/LLM models significantly more reliable.

11. Conclusion

The rapid growth in applicant volume and the diversity of student CV formats have made traditional recruitment workflows increasingly difficult to manage in large academic institutions. This project addresses these challenges by developing a set of four independent yet complementary AI systems designed to improve the accuracy, speed, transparency, and privacy of CV screening and institutional information retrieval.

The first system, a RAPTOR-based campus chatbot (KGPGPT), demonstrates the effectiveness of hierarchical RAG techniques for extracting reliable answers from long, complex institutional documents. The second system, a Multi-Agentic CV Screening Dashboard, introduces a structured agent-based approach to semantic JD–CV matching, enabling fast and explainable candidate filtering. The third subsystem—a Hybrid Agentic Shortlisting Framework—combines deterministic scoring with LLM-driven reasoning to produce fair and auditable rankings suitable for large applicant pools. Finally, the Fully Local Skills Evaluation Model offers a privacy-preserving, CPU-only solution by integrating classical NLP, embeddings, and lightweight local LLMs for offline skill relevance assessment.

Together, these systems showcase how a blend of modern NLP techniques, multi-agent architectures, hybrid scoring models, and local LLM pipelines can address the limitations of keyword-based ATS systems and high-cost cloud LLM solutions. Each subsystem contributes unique strengths: improved retrieval accuracy, scalable semantic screening, explainable scoring, and complete data privacy.

Although constraints in data availability and computational resources limited the deployment of larger and more capable models, the results indicate that even lightweight architectures can provide meaningful improvements in recruitment workflows. The work establishes a strong foundation for future advancements such as RAPTOR-enabled CV parsing, ML-augmented LLM scoring, enhanced multi-agent designs, and the development of high-quality labeled datasets.

Overall, this project demonstrates a practical and technically robust approach to modernizing campus recruitment processes. It highlights the potential of AI-driven systems to deliver fast, fair, and institution-friendly solutions in environments where accuracy, transparency, and privacy are of paramount importance.